Week 7 Tutorial: Causality

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Plan for today

- No-coding
- Three approaches to causal inference
  - Natural Experiments
  - Survey experiments
  - Field Experiments

Plan for each paper

Research Question → Research Design → Outcome Interpretation

Before we start…

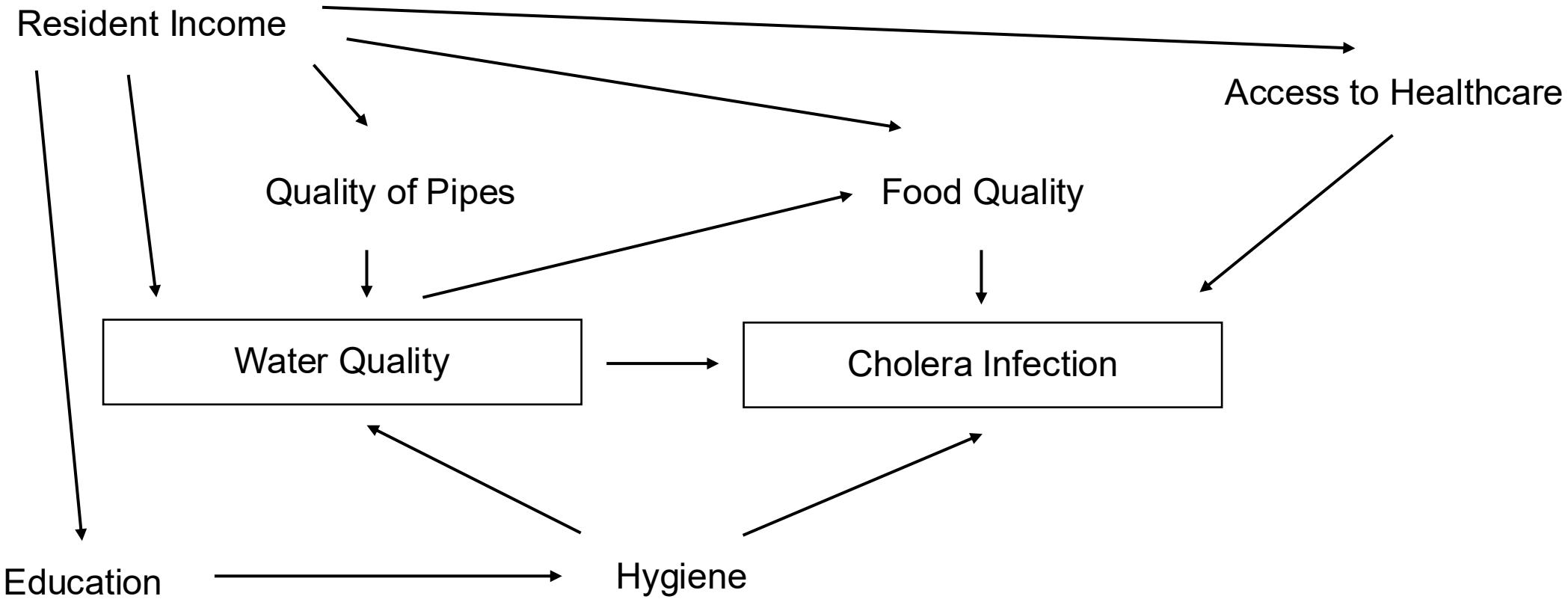
- What makes a good causal inference?
- What will we want to see in the papers?
- How can we evaluate the findings?

Paper 1 – Natural Experiments (John Snow, 1855)

- Two competing theories about the causes of cholera
  - Transmission by air (“miasmas”)
  - Transmission by water (germs)
- Snow identified clusters of cholera outbreaks next to water pumps (figure 1 in handout materials)

- Let’s construct a **DAG for the germs theory**
  - What is the outcome?
  - What is the predictor?
  - What are the **front door** and the **back door** paths?

Paper 1 – Natural Experiments (John Snow, 1855)



Paper 1 – Natural Experiments (John Snow, 1855)

- To study the impact of water quality on cholera, Snow exploits a **natural experiment**
- In 1854, there were three major water providers in London
  - The Southward and Vauxhall Company used water from part of Thames contaminated with sewage
  - The Lambeth Water Company pumped its water from the part of Thames that was not contaminated

- How can this variation be exploited for the purposes of **causal identification**?

Paper 1 – Natural Experiments (John Snow, 1855)

- Snow collected data on each death during the 1854 cholera epidemic, and identified the water supplier for the house of the deceased
- Below are the **results** of his research:

Table IX of Snow (1855):

Water Company	Houses	Deaths	Deaths per 10000 Houses
Southwark & Vauxhall	40,046	1,263	315
Lambeth	26,107	98	37
Rest of London	256,423	1,422	59

Paper 1 – Natural Experiments (John Snow, 1855)

- What can we conclude from the figures?
- Under what **assumptions** does the research identify a **causal relationship**?

- Snow implements a version of a **difference-in-difference design**:
  - There were cholera epidemics in 1849 and 1853
  - Until 1852, Lambeth used contaminated water source
  - Snow compares the change in the number of deaths between the two epidemics between the two companies

- Does this research design make the inference stronger and why?

Paper 1 – Natural Experiments (John Snow, 1855)

- For the purposes of causal inference analysis…
  - What are the **strengths** of the natural experiments approach?
  - What are the **weaknesses**?

Paper 2 – Survey Experiments (Charnysh et al., 2014)

- Charnysh and co-authors study the impact of **national identity salience** on **pro-social behaviour**

- Imagine you have the resources to run your own study
  - What would be your **outcome variable**?
  - What would be your **treatment**?
  - How would a **causal diagram** look like?

Paper 2 – Survey Experiments (Charnysh et al., 2014)

- The study conditions on core demographic characteristics
  - What problem does it address?

Table 1. Demographic Characteristics of the Sample.

	R_t	R_c	M_t	M_c	India
Age	28.38	28.42	27.12	29.37	.25
Male	0.62	0.66	0.62	0.66	.52
College graduate	0.92	0.88	0.91	0.90	.21
Monthly income > Rs. 20,000	0.36	0.40	0.36	0.40	.6
Urban	0.51	0.51	0.51	0.50	.28
Upper caste	0.45	0.43	0.43	0.41	.32
Observations (*n*)	231	221	239	227	

Paper 2 – Survey Experiments (Charnysh et al., 2014)

- Charnysh and co-authors run a **list experiment** in India
  - **Treatment:** increased salience of shared (Indian) national identity
  - **Outcome:** donations by members of a dominant ethnic group (Hindus) to members of a rival, minority group (Muslims)

- Read extract 2.1 in the handout materials
  - How is the treatment constructed?
  - How can you assess the **construct validity**?

Paper 2 – Survey Experiments (Charnysh et al., 2014)

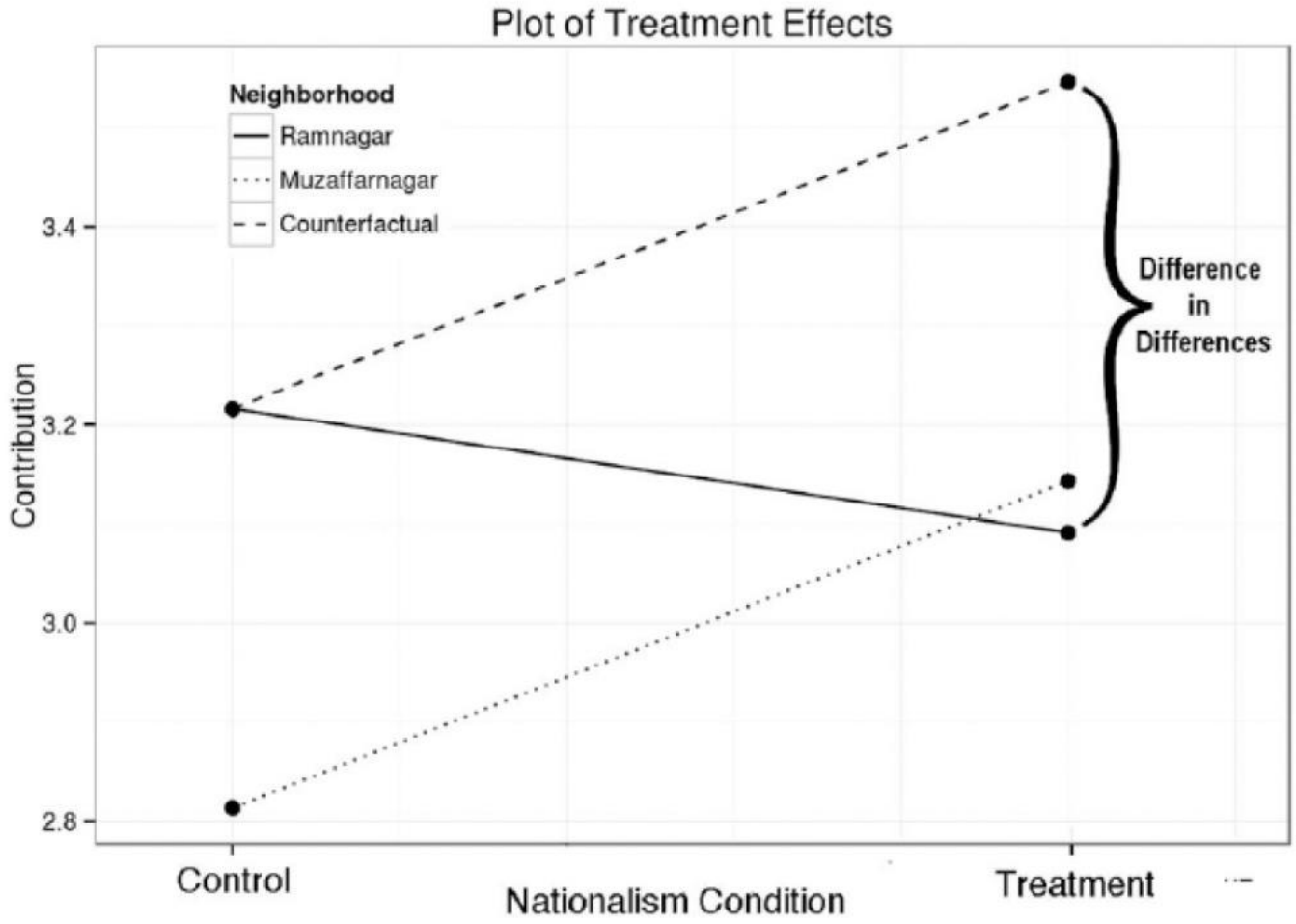
- Four groups – four outcomes

	Treatment group	Control group
Muslim-majority city (Muzaffarnagar)	Y_{mt}	Y_{mc}
Hindu-majority city (Ramnagar)	Y_{ht}	Y_{hc}

Paper 2 – Survey Experiments (Charnysh et al., 2014)

- The results of the study are presented in Table 2.2 of the handout materials for paper 2
  - What is the effect of the treatment for the **control group**?
  - What is the effect of the treatment for the **treatment group**?

Paper 2 – Survey Experiments (Charnysh et al., 2014)



Paper 2 – Survey Experiments (Charnysh et al., 2014)

- For the purposes of causal inference analysis…
  - What are the **strengths** of the survey experiments approach?
  - What are the **weaknesses**?

Paper 3 – Field Experiments (Gerber, Karlan, Bergan, 2009)

- Gerber, Karlan, and Bergan study the impact of **media exposure** on **political behaviour and voting** in the US

- Assume that we have data on **media subscriptions** (treatment) and **voting** (outcome) at individual level
  - Does this data allow us to identify the causal effect of media exposure?
  - What additional data and assumptions would we require to identify a causal relationship?
- In terms of the DAG, what are the **front paths** and the **backdoor paths**?

Paper 3 – Field Experiments (Gerber, Karlan, Bergan, 2009)

- Gerber, Karlan, and Bergan implement a **field experiment** for 2005 gubernatorial elections in Virginia

- Read extract 3.1 in the handout materials
  - How is the treatment constructed?
  - How does it help to isolate the front door paths?

Paper 3 – Field Experiments (Gerber, Karlan, Bergan, 2009)

- Two treatments – three groups

Washington Post	New York Times	Control
Y_{WP}	Y_{NYT}	Y_C

Paper 3 – Field Experiments (Gerber, Karlan, Bergan, 2009)

- The results of the OLS regression are presented in table 3.2 in the handout materials
  - What is the effect of the treatment on **voting** in gubernatorial elections?
  - What is the effect of the treatment on **candidate selection**?

- Gerber, Karlan, and Bergan include covariates in their specification
  - What are the arguments in favour of including covariates?
  - What are the arguments in against including covariates?

Paper 3 – Field Experiments (Gerber, Karlan, Bergan, 2009)

- For the purposes of causal inference analysis…
  - What are the **strengths** of the field experiments approach?
  - What are the **weaknesses**?

Final thoughts

- We covered three types of **causal identification strategies**:
  - Natural experiments
  - Survey experiments
  - Field experiments

- What approach did you find the most **convincing** and why?

- What approach is the most **relevant** to your research?

References

- Snow, J., 1855. *On the mode of communication of cholera*. 2nd ed., much enlarged. London: John Churchill.
- Charnysh, V., Lucas, C. and Singh, P., 2015. The ties that bind: national identity salience and pro-social behavior toward the ethnic other. *Comparative Political Studies*, 48(3), pp.267–300.
- Gerber, A.S., Karlan, D. and Bergan, D., 2009. Does the media matter? A field experiment measuring the effect of newspapers on voting behavior and political opinions. *American Economic Journal: Applied Economics*, 1(2), pp.35–52.